

Self-attention learning network for face super-resolution

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ABSTRACT

Existing face super-resolution methods depend on deep convolutional networks (DCN) to recover high-quality reconstructed images. They either acquire information in a single space by designing complex models for direct reconstruction, or employ additional networks to extract multiple prior information to enhance the representation of features. However, existing methods are still challenging to perform well due to the inability to learn complete and uniform representations. To this end, we propose a self-attention learning network (SLNet) for three-stage face super-resolution, which fully explores the interdependence of low- and high-level spaces to achieve compensation of the information used for reconstruction. Firstly, SLNet uses a hierarchical feature learning framework to obtain shallow information in the low-level space. Then, the shallow information with cumulative errors due to DCN is improved under high-resolution (HR) supervision, while bringing an intermediate reconstruction result and a powerful intermediate benchmark. Finally, the improved feature representation is further enhanced in high-level space by a multi-scale context-aware encoder-decoder for facial reconstruction. The features in both spaces are explored progressively from coarse to fine reconstruction information. The experimental results show that SLNet has a competitive performance compared to the state-of-the-art methods.

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1. Introduction

Face images have a unique distribution in terms of their highly structured shapes and rich contextual information compared to other real-world images. It is gradually becoming one of the critical research objects in image processing, computer vision and other fields. In addition, advanced face applications (e.g., face recognition, face manipulation and face alignment) are becoming increasingly sophisticated, and as a result, many face systems are widely used (Li, Tu, & Xu, 2022; Wang et al., 2021). Nevertheless, face images are suffered from inferior imaging equipment and a complex imaging environment, which leads to low-resolution captured images and makes face systems difficult. Consequently, it is of practical value to study this pre-processing technique for low-resolution (LR) face images to meet the needs of existing face recognition systems. Face super-resolution (FSR) is a technique to recover LR images into HR images with faithful visual quality, and it is widely used in criminal investigation and security surveillance.

FSR, an ill-posed problem, should fully use its own valid and relevant information for reconstruction. Traditional methods try to learn the correspondence between LR and HR feature spaces through many representative example pairs, such as neighbor embedding, sparse coding and random forest. Although these methods are computationally efficiency and converge quickly, they tend to lose high-frequency detail information, limiting the performance of the reconstruction, especially in the case of a large scaling factor.

With the development and wide application of deep learning techniques, FSR methods based on convolutional neural networks (CNNs) are becoming more and more popular and significantly outperform traditional methods. One of the most intuitive solutions is to design deep convolutional networks with powerful representational learning capabilities for the purpose of restoring low-resolution images. On the one hand, some approaches investigate supervised learning methods based on PSNR and GAN, such as SRCNN (Dong, Loy, He, & Tang, 2016), VDSR (Kim, Lee, & Lee, 2016), PUSLE (Menon, Damian, Hu, Ravi, & Rudin, 2020), ESRGCNN (Tian et al., 2022), which more or less use pixel-wise error loss functions and the main part of the network is a stacked residual block structure. Therefore, as the network

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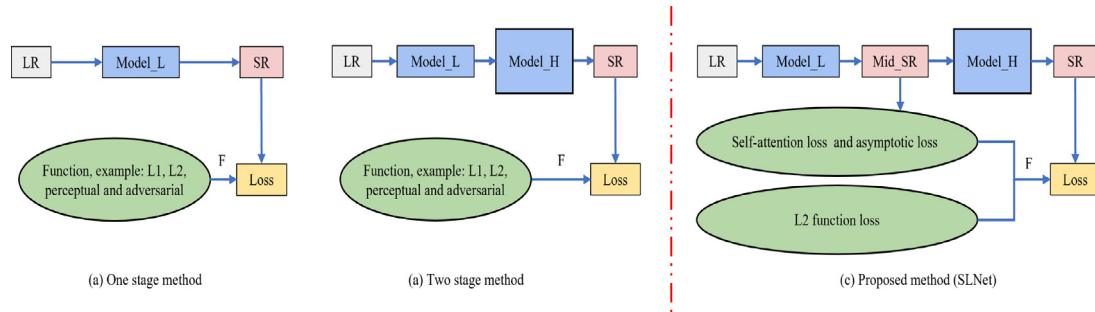


Fig. 1. (a) The one-stage-based SR method reconstructs SR directly from LR, which uses loss functions such as L1, L2 and perceptual loss. (b) The two-stage SR method explores the reconstruction information in LR and HR feature space separately, which also uses the loss functions such as L1, L2 and perceptual loss. (c) The proposed self-attention learning network (SLNet) continuously improves the reconstruction performance by self-attention learning based on the three-stage approach. Due to the intermediate results generated, self-attention loss, asymptotic loss and reconstruction loss are used as the loss functions of SLNet. Note that Model_H and Model_L denote feature learning networks in HR and LR spaces, respectively.

deepens, it can extract rich facial features, which also extracts too much redundant feature information. As a result, it yields poor detail reconstruction, blurred edge contours, failure to recover high-frequency detail feature information of facial components, and poor visual perception.

On the other hand, with the help of unique feature information of face images, such as identity information and facial formation information, many methods incorporate face attribute information into super-resolution reconstruction. [Chen, Tai, Liu, Shen, and Yang \(2018\)](#) demonstrated the effectiveness of prior information for image recovery, and the more prior information, the better the quality of the recovered image. [Ma, Jiang, Rao, Lu, and Zhou \(2020\)](#) designed an iterative feedback network so that the generated super-resolution images increasingly converge to the real images. The more realistic images make it possible to extract more accurate a priori information through the a priori network to improve the quality of the images again. [Yu, Zhang, and Xie \(2021\)](#) proposed a semantic-driven residual network based on a generative adversarial network to restore the HR face image with proper identification from the LR face image. Although they achieve good performance, when extracting the prior information and fusing it into the super-resolution backbone network, irrelevant information is inevitably introduced or generated, which weakens the constraint of the prior information and leads to insufficient texture details in the generated face images.

To address the above issues, we propose a self-attention learning network. It is a novel end-to-end three-stage learning framework that can fully exploit the mutually compensated feature information in low-level and high-level spaces to enhance the representational capabilities. In contrast to existing multi-stage methods ([Fan et al., 2017](#); [Hui, Wang, & Gao, 2018](#); [Wang, Zhong, Jiang, Zhai, & Liu, 2020](#)), the proposed approach uses self-attention learning to explore the association between high-level and low-level semantic features and progressively improves the performance of the reconstruction by refining super-resolution network rather than directly using the deep learned features for reconstruction (see [Fig. 1](#)). First, the low-level face super-resolution network is designed to learn shallow features of the face. We developed a hierarchical feature learning module with RDB as the basic unit to extract shallow features that are conducive to fusion with higher-level features. It not only compensates for the loss of feature information during convolutional propagation and increases the perceptual field information, but also enables the network to learn more detailed features of the face, thus improving the network's ability to reconstruct local texture regions of facial images. Then, the self-attention module uses the output features of LFSR (Low-Level Face Super-resolution) and the LR face image as input, and generates an attention map through attention-based supervised learning to

suppress the redundant features of LFSR, allowing only useful features to be propagated to the HFSR (High-Level Face Super-resolution) stage. Finally, HFSR enhances multi-scale contextual information by using an encoder-decoder with multiple attention units embedded for the final reconstruction. Experimental results show that our approach achieves state-of-the-art face hallucination performance.

In summary, this paper has the following contributions.

- We propose a three-stage approach for face super-resolution that continuously explores valuable information in the HR-LR space to capture powerful face representations.
- An efficient self-attention module is proposed, which progressively enhances the representation of features from the low-level space to the high-level space in a self-supervised manner, thereby improving the redundant information introduced when the features are propagated forward.
- We demonstrate the effectiveness of the proposed method, which shows competitive performance for face super-resolution on two publicly available datasets and one real-world dataset. In addition, we provide detailed ablation experiments, qualitative results and generalization tests.

The rest of the paper is organized as follows. Section 2 reviews the related work in image SR and face SR, Section 3 provides the details of our method, Section 4 describes our implementation and results, and Section 5 concludes the whole paper.

2. Related work

In recent years, numerous single-image super-resolution (SISR) methods have been proposed in the field of computer vision. In this section, we briefly review the deep learning-based SR methods, which are close to our approach.

2.1. Image super-resolution

SRCNN ([Dong, Loy, He, et al., 2016](#)), as a pioneering work of deep learning in super-resolution, is a three-layer convolutional network structure, including feature extraction, nonlinear mapping and reconstruction. [Kim et al. \(2016\)](#) used deep networks to recover SR images from bicubic interpolation of the input LR images. Taking advantage of the architecture of deep residual learning ([He, Zhang, Ren, & Sun, 2016](#)), [Lim, Son, Kim, Nah, and Lee \(2017\)](#) proposed an enhanced deep super-resolution network by using improved residual blocks without batch normalization. [Harris, Shakhnarovich, and Ukita \(2018\)](#) combined iterative up-sampling and downsampling layers to recover SR images. With the help of residual learning, [Zhang, Tian, Kong, Zhong, and Fu](#)

(2018) proposed RDB to leverage multi-level features. Ge, Gong, and Yu (2018) recovered the SR image from the LR image by concatenating the output features using a network structure of several residual dense blocks (RDBs). In order to strengthen the relationship between intermediate layer features, Dai, Cai, Zhang, Xia, and Zhang (2019) proposed a second-order attention network to achieve stronger feature representation and feature relationship learning. By considering the relationship between spatial and channel dimensions, Zhang, Li, Li, Wang, Zhong, and Fu (2018) proposed a very deep residual channel attention network (RCAN) for high precision image SR. To make full use of spatial-temporal information, Yi et al. (2022) proposed a progressive fusion network for video SR that processes frames in a progressive separation and fusion manner. Additionally, Ledig et al. (2017) used a perceptual loss function for generating natural-looking images with high perceptual quality (Yang et al., 2022). Wang et al. (2018) presented an enhanced SRGAN, which uses residual densely connected blocks (Huang, Liu, van der Maaten, & Weinberger, 2017).

2.2. Face super-resolution

Face super-resolution, also known as face hallucination (FH), is a particular case of the image super-resolution field. FH is pioneered by Baker and Kanade (2002), who demonstrated that it is possible to perform SR with high magnification on specific categories of images (e.g. faces and text). Zhu, Liu, Loy, and Tang (2016) proposed a cascaded two-branch network to optimize face hallucination and dense correspondence field estimation within a unified framework. They further refined their model to handle unaligned faces (Yu, & Porikli, 2017a), noisy faces (Yu, & Porikli, 2017b), and faces with different attributes (Yu, Fernando, Hartley, & Porikli, 2020). Xin, Li, Jiang, Wang, and Gao (2022) proposed to reconstruct HR images by predicting wavelet coefficients from LR images rather than directly inferring HR face images. Considering the unique structural attributes of faces, Song, Zhang, He, Bao, and Yang (2017) learned face component generation and enhancement (LCGE) to achieve face image reconstruction. Wang, Lu, Xu, and Zhang (2020) proposed a face super-resolution method using multi-view texture compensation. In addition, Chen, Gong, Wang, Li, and Wong (2021) introduced a new spatial attention residual network (SPARNet), which is built on face attention units (FAUS), for face super-resolution. Lu et al. (2021) proposed a split-attention in split-attention network (SISN) to fuse internal channel features and external multi-path features for exploring face structure information.

The latest work employs additional face a priori supervision, such as face resolution maps (Chen et al., 2018), landmark heat maps (Bulat & Tzimiropoulos, 2018; Yu, Fernando, Ghanem, Porikli, & Hartley, 2018; Zhuang, Li, Zhang, Li, & Lu, 2022), and identity information (Zhang, Zhang, Cheng, Hsu, Qiao, & Liu, 2018), to train their networks. Kim, Kim, Kwon, and Kim (2019) proposed a facial attention loss that focuses the network on landmark regions. Ma et al. (2020) introduced an iterative approach to iteratively predict SR results and landmarks. In addition, Jiang et al. (2020) used adaptive thresholding to fuse multiple SR models to exploit their advantages.

Although the additional supervision helps to improve performance, not only does this take extra time to tag the data, but it is an intermediate way of directing the SR network to fundamental facial structures. In contrast to these approaches, without additional constraints, we use a self-attention learning module to constrain the intermediate results generated in the first stage and to bridge the first and second stage features to forward propagation, enhancing the complementarity and representation of information.

3. Method

In this section, we first describe the overall architecture of our method. Then, the main components of our method are described, including the low-level space FSR network, high-level space FSR network and the used various components. Finally, the loss function is discussed in detail.

3.1. Network architecture

The overall structure of our proposed convolutional network for learning self-attention features is shown in Fig. 2. The basis of our method consists of two cascaded sub-networks and a self-attention module. The first sub-network is the low-level face super-resolution network, which aims to generate an initial super-resolution result for the input low-resolution image using hierarchical features. The second sub-network, named the high-level face super-resolution network, improves the results of the first sub-network by exploiting the attention-based and encoder-decoder subnetworks to take advantage of the wide range of contextual information available from large perceptual fields. As a self-attention module connecting the two cascaded sub-networks, it bridges the low-level space and high-level space features to achieve mutual compensation of information, progressively improving the clarity of the low-resolution image.

Suppose I_{LR} and I_{SR} represent the input LR image and the reconstructed image, respectively. Firstly, we describe the initial LR input as a set of feature maps H_0 by applying a standard convolutional layer,

$$H_0 = f_{init}(I_{LR}), \quad (1)$$

where $f_{init}(\bullet)$ indicates the initialization of the convolution operation.

Then, the feature H_0 is used in the low-level space face super-resolution sub-network, which extracts the hierarchical features in the low-resolution,

$$H_{LF} = f_{LFSR}(H_0), \quad (2)$$

where $f_{LFSR}(\bullet)$ denotes the function of hierarchical feature learning block (HFLB) to reuse multi-layer LR features.

Immediately afterward, the extracted feature H_{LF} is fed to the self-attention module to capture the high-frequency representation H_{mid} of the low-resolution space, as well as to generate intermediate super-resolution result I_{mid} , by

$$H_{mid}, I_{mid} = f_{SAM}(H_0, H_{LF}), \quad (3)$$

where $f_{SAM}(\bullet)$ is the function of self-attention module (SAM). Note that the SAM contains the upsampling process.

Finally, we obtain H_{HF} by feeding H_{mid} into the HFSR sub-network, which uses the encoder-decoder subnetwork to learn multi-scale features in a skip-connection approach. Specifically,

$$H_{HF} = f_{HFSR}(H_{mid}), \quad (4)$$

where $f_{HFSR}(\bullet)$ is high-level space face super-resolution network.

As a result, the final output I_{SR} can be obtained after the final convolutional layer,

$$I_{SR} = f_{final}(H_{HF}), \quad (5)$$

where $f_{final}(\bullet)$ denotes the final convolution layer.

3.2. Low-level space FSR network

Involving more information that facilitates reconstruction and provides a high starting point for subsequent subnetworks, we develop a HFLB based on a residual dense network (RDN) (Zhang,

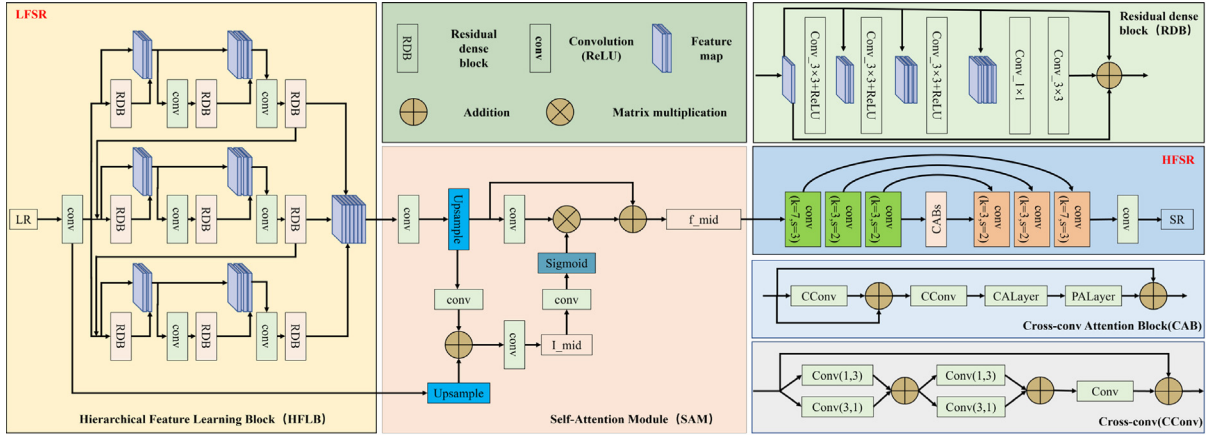


Fig. 2. Proposed self-attention feature learning architecture for face super-resolution. The first stage adopts hierarchical feature learning clusters to extract local low-level features. The second stage refines the features output from the first stage by building a self-attention module and feeding it to the third stage. The third stage utilizes an attention-based encoder–decoder in high-resolution space to learn multi-scale contextual fine-grained features to achieve the final face reconstruction.

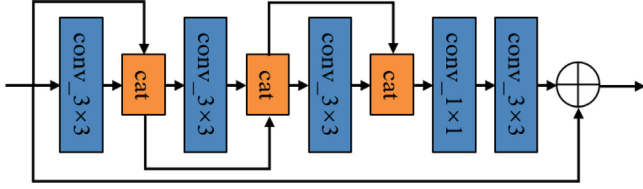


Fig. 3. Illustration of the structure of the residual dense module, which contains multiple densely connected convolutional layers.

Tian, et al., 2018), which learns hierarchical features to achieve super-resolution in the first step.

As we know, RDN is one of the most advanced methods for image SR. For details of the residual dense block (RDB) are shown in Fig. 3. Therefore, the output of the RDB layers is

$$\begin{aligned} C_i &= f_{conv}^i(H_{i-1}), s.t. i = 1, 2, 3, 4, 5 \\ H_i &= \text{cat}(H_{i-1}, C_i), s.t. i = 1, 2, 3, \dots \\ H_{RDB} &= C_5 \end{aligned} \quad (6)$$

where $f_{conv}^i(\bullet)$ is the i th convolution function in the RDB. C_i and H_i denote the output after the i th Convolution and the i th Concatenate, respectively. Note that the kernel size is 3×3 for all four convolution layers except for the fourth convolution layer, which has a kernel size of 1×1 . The output of the RDB is the output of the last convolutional layer.

For the sake of improving the capability of feature learning, we design a three-path network with hierarchical feature awareness using RDB as the basic unit by the idea of a feedback scheme, where the schematic diagram of a single path is shown in Fig. 4.

Unlike the usual recurrent structure, each path accepts shallow-level features as part of the input and sends the output to subsequent modules. In each path, we use shallow features H_0 of the LR image as part of the input to provide low-level information. First, the shallow features are refined by the hidden state H_{RDB}^{t-1} of the previous iteration as

$$R_0 = f_{conv}(\text{cat}(H_0, H_{RDB}^{t-1})), \quad (7)$$

where R_0 is the refined feature map. Then, R_0 is sent to multiple densely connected RDB groups (RDBGs), which represent each path. The densely connected structure enables the model to incorporate multi-level features and more efficient information propagation. We progressively collect the output of all RDBGs through feature concatenation. To avoid the problem of high

computational complexity caused by repeated concatenation, we adopt a 1×1 convolution before each RDBG to compress the concatenated feature and reduce the number of channels. To be precise, the output of the d th RDBG is computed as

$$R_d = f_{RDB}(f_{conv}(\text{cat}(R_0, R_1, R_{d-1}))). \quad (8)$$

We regard the output of the last RDBG as the hidden state of the current iteration, i.e., $H^t = R_D$, where D is the number of RDBGs in each path and t denotes the t th path. Finally, we incorporate a global skip connection in each iteration to allow the original information to bypass the subnetwork. The output of the t th iteration F_t is computed as

$$H_{LF} = \text{cat}(H^1, H^2, \dots, H^t). \quad (9)$$

Afterwards, we then feed the joint hierarchical low-level features to the self-attention module.

3.3. Self-attention module

Recent multi-stage networks for super-resolution (Fan et al., 2017; Hui et al., 2018) directly predict an image at each stage, which is then passed to the next consecutive stage. Instead, we introduce a self-supervised attention module between two stages, which facilitates achieving significant performance gain. The schematic diagram of SAM is shown in Fig. 5, and its contributions are two-fold. First, it provides ground-truth supervisory signals useful for the two stages of progressive image super-resolution. Second, with the help of locally supervised predictions, we generate attention maps to suppress the less informative features of the earlier stages and only allow the useful ones to propagate to the later stage.

As illustrated in Fig. 5, SAM takes the incoming features H_{LF} of the earlier stage and first generates a residual image with a simple 1×1 convolution and upsampling operation. The residual image is added to the upsampled low-resolution input image to obtain the pre-reconstructed image I_{mid} . To this predicted image I_{mid} , we provide explicit supervision with the ground-truth image. Next, per-pixel attention masks F_{mask} are generated from the image I_{mid} using a 1×1 convolution followed by the sigmoid activation. These masks are then used to re-calibrate the transformed local features (obtained after 1×1 convolution), resulting in attention-guided features which are added to the identity mapping path. Finally, the attention-augmented feature representation H_{mid} , produced by SAM, is passed to the next stage for further processing.

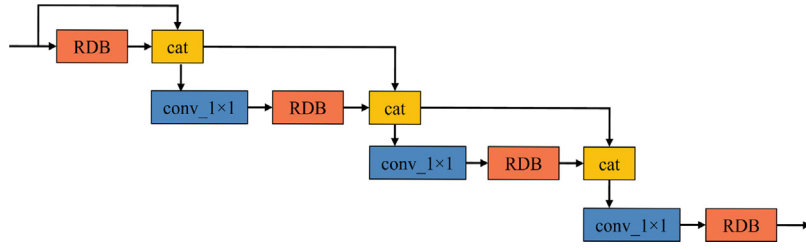


Fig. 4. The architecture of a single-path hierarchical feature learning module which consists of multiple densely connected RDBs.

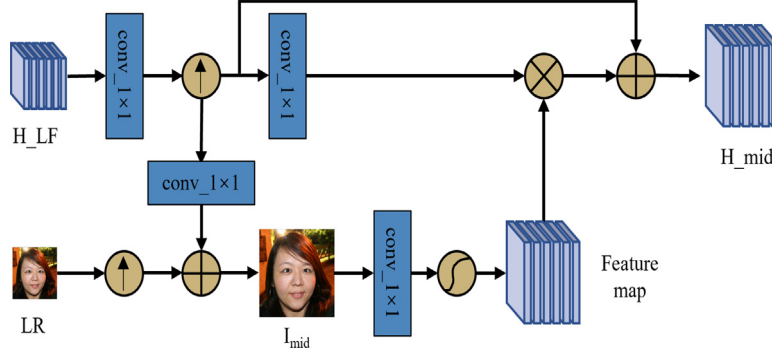


Fig. 5. A diagram of the structure of the self-attention module.

3.4. High-level space FSR network

Super-resolution reconstruction of HR images from LR images requires the simultaneous recovery of accurate global appearance and visually friendly details such as textures and sharp edges. Inspired by [Ronneberger, Fischer, and Brox \(2015\)](#), we use an encoder–decoder framework with skip connections to extract multi-scale convolutional features. The detailed network structure is shown in the right part of [Fig. 2](#). The encoder uses three convolution layers to extract multi-scale features from the input, which are forwarded to the decoder via a skip connection. Note that two different convolution operations are used here, a 7×7 and a 3×3 convolution kernel. The decoder has an almost symmetric structure with the encoder, which uses a deconvolution layer, again using the two different convolution kernels above, to explore higher-level features by refining the multi-scale features. Furthermore, with the advantage of cross-convolution ([Liu et al., 2022](#)), we use groups of cross-convolutional attention block (CABs) to connect the encoder and decoder. Specifically, we obtain H_{HF} by the following equation,

$$H_{HF} = f_{Decoder}(f_{CABs}(f_{Encoder}(H_{mid}))), \quad (10)$$

where $f_{Encoder}$, $f_{Decoder}$, f_{CABs} denotes the function of the encoder, the function of the decoder and the function of the groups of cross-convolutional attention block, respectively.

Such an operation allows our network to focus more on the attribute information of the face to recover local details. In the experimental section, we will focus on the benefits of these components.

3.5. Network training

In this section, we will discuss the optimization function of the proposed method. Several loss functions have been investigated, such as L_2 ([Dong, Loy, He, & Tang, 2014](#); [Dong, Loy, & Tang, 2016](#); [Kim et al., 2016](#); [Tai, Yang, & Liu, 2017](#); [Tai, Yang, Liu, & Xu, 2017](#); [Wang, Wang, Wang, Li, & Lu, 2020](#)), L_1 ([Lai, Huang, Ahuja, & Yang, 2017](#); [Lim et al., 2017](#); [Zhang, Tian, et al., 2018](#)), perceptual and adversarial loss ([Ledig et al., 2017](#)). As for our method,

we optimize the same loss function as previous works (e.g., L_2 loss function) by minimizing the difference between the two images. Following the proposed method involving two stages, the total loss function designed consists of three components: reconstruction loss L_{rec} , self-attention loss L_{sam} and asymptotic loss L_{asy} . Here, L_{asy} denotes the error between the middle reconstruction result after SAM output and the final result after HFSR, so as to constrain the HFSR network to further improve the reconstruction performance.

Given a training dataset with N image pairs $\{I_i^{LR}, I_i^{HR}\}_{i=1}^N$, the objective function of our method can be written as

$$\begin{aligned} Loss &= \varpi_1 \times L_{rec} + \varpi_2 \times L_{sam} + \varpi_3 \times L_{asy} \\ &= \varpi_1 \times \frac{1}{N} \sum_{i=1}^N \|I_{SR}^i - I_{HR}^i\|_2 + \varpi_2 \times \frac{1}{N} \sum_{i=1}^N \|I_{mid}^i - I_{HR}^i\|_2 \\ &\quad + \varpi_3 \times \frac{1}{N} \sum_{i=1}^N \|I_{mid}^i - I_{SR}^i\|_2, \end{aligned} \quad (11)$$

where ϖ_1 , ϖ_2 and ϖ_3 denote the weighting factors for each of the three losses.

4. Experimental results

In this section, we first introduce our experimental data and implementation details. Then, we compare the proposed approach with several state-of-the-art methods from different perspectives. Finally, we present ablation studies to investigate our network.

4.1. Datasets

Flickr-Faces-High-Quality (FFHQ) ([Karras, Laine, & Aila, 2019](#)) is a high-quality face dataset containing 70,000 HD face images in PNG format at 1024×1024 resolution, with a wide variety and variation in age, race and image background, as well as a wide range of face attributes. FFHQ images are crawled from Flickr, aligned and cropped using Dlib, and then algorithmically removed from non-real faces such as statues, paintings and photographs. Some sample images are shown in the top half of [Fig. 6](#). Here, we

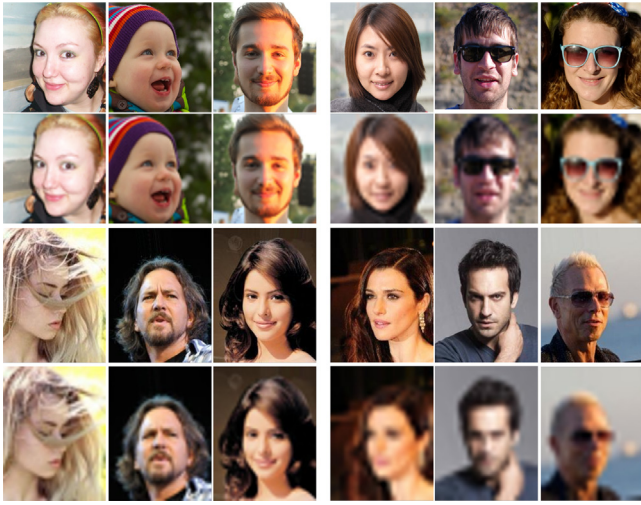


Fig. 6. Some sample images from the FFHQ (upper two rows) and CelebA (bottom two rows) datasets, which incorporate the effects of lighting, angle and mask. Note that the odd-numbered rows indicate HR and the even-numbered rows indicate LR.

resize the original image to 256×256 as HR and then get the corresponding LR (64×64 and 32×32) with sampling factors 4 and 8, respectively.

CelebA (Liu, Luo, Wang, & Tang, 2015) is a large in-the-wild database that contains more than 200,000 face images under different poses, occlusions and background variations. Some sample images are shown in the bottom half of Fig. 6. First, MTCNN (Zhang, Zhang, Li, & Qiao, 2016) is used to detect faces and crop out face regions without any alignment operation. Then, images are resized to 200×160 by bicubic interpolation, which is defined as the HR image. Finally, the LR images 50×40 (25×20) is obtained by downsampling the HR image with a factor of 4 (and 8).

LDHF (Maeng, Liao, Kang, Lee, & Jain, 2012) dataset is acquired with VIS and NIR images of the same target at different distances (1 m, 60 m, 100 m and 150 m) with 100 subjects. Some sample images are shown in Fig. 7. Here, we only use the images taken at 1 m and 150 m for validation. Note that MTCNN (Zhang et al., 2016) is used to crop out the face region with a size of 200×160 (the same image size as the CelebA dataset). Since the distance is different, the resolution of the image is also different. Therefore, the cropped face image at a far distance is used as an input and the cropped face image at a near distance is used as a reference comparison to verify the performance of different methods.

4.2. Implementation details

To avoid overfitting, we increased the diversity of the training samples in our experiments by the following operations: no manipulation (i.e., the original image), 90° , 180° , 270° random rotation, and horizontal flip. In addition, we investigated the basic network parameters: the number of RDB, RDBG and CAB (denoted as K , G , M respectively) in SLNet.

Our method was implemented in PyTorch on a PC with two Nvidia RTX 2080Ti GPUs. The batch size was set to 32 and the learning rate was initially set to 1×10^{-4} . The training was stopped after 5×10^4 epochs. For quantitative comparison, all our experiments used PSNR, SSIM (Wang, Bovik, Sheikh, & Simoncelli, 2004), FSIM and VIF (Sheikh & Bovik, 2006) as metrics for performance evaluation.

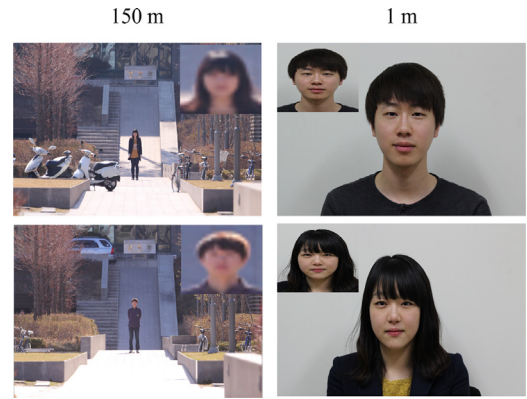


Fig. 7. Two examples of the original images from the LDHF dataset and the cropped face regions, where the original images on the left and the right are taken by the camera at 150 m and 1 m from the person, respectively. Different distances lead to different quality of the captured images. Additionally, MTCNN (Zhang et al., 2016) is used to crop the face region.

4.3. Comparison to the state-of-the-arts

We compare our method with several state-of-the-art methods including DBPN (Haris et al., 2018), RDN (Zhang, Tian, et al., 2018), SAN (Dai et al., 2019), RCAN (Zhang, Li et al., 2018), SRFBN (Li et al., 2019), MTC (Wang, Lu, et al., 2020), SPARNet (Chen et al., 2021), SISR (Lu et al., 2021). For fair comparison, we retrain all comparison methods on the same training datasets as ours. We also use bicubic interpolation as a baseline method. To thoroughly verify the validity of the proposed method, we illustrate the results in the following aspects.

4.3.1. Analysis of objective indicators

Comparison of the objective indicators for the different methods are listed in Table 1. Our method achieves very competitive PSNR, SSIM, FSIM, and VIF on FFHQ and CelebA. Specifically, SLNet achieves the best or second best performance on all metrics. For FFHQ, SLNet's PSNR, SSIM, FSIM, and VIF are only lower than SISR (Lu et al., 2021) at scale of 4, while SLNet achieves the highest metrics at scale of 8, except for VIF. For CelebA, SLNet achieved the best metrics at scale of 4 and 8, except for SSIM at scale of 8. Furthermore, we note that SISR (Lu et al., 2021) and SPARNet (Chen et al., 2021) are very close to the performance of SLNet in all metrics. SPARNet (Chen et al., 2021) utilizes a novel spatial attention residual network that can adaptively guide features related to critical face structures and focus less on those regions with fewer features. SISR (Lu et al., 2021) utilizes an external-internal split attention group (ESAG) is responsible for both facial structure information and facial texture details. Notably, the improvement of our method in various metrics is significant on the CelebA dataset. CelebA is a database collected in the wild with relatively low-quality images under significant pose, illumination, and expression variations, so it has a relatively large baseline and significant variance. Unlike the above methods of directly extracting reconstructed information, SLNet not only utilizes a dual-space network to gradually recover structural information and enhance textures, but also designs self-attention learning module to indirectly deal with the problem of information discrepancy between dual spaces, and also maintains good performance for data with small baselines.

4.3.2. Analysis of subjective quality

Figs. 8 and 9 show the qualitative results of SR at scale of 8 on the two datasets, respectively. Compared to state-of-the-art super-resolution methods, our method faithfully recovers face

Table 1

Comparison of the super-resolution performance of the different methods. The best and second best are marked in red and blue respectively.

Dataset	Scale	Evaluation	Bicubic	DBPN (Haris et al., 2018)	RDN (Zhang, Tian, et al., 2018)	SAN (Dai et al., 2019)	RCAN (Zhang, Li et al., 2018)	SRFBN (Li et al., 2019)	MTC (Wang, Lu, et al., 2020)	SPARNet (Chen et al., 2021)	SISN (Lu et al., 2021)	SLNet
FFHQ	×4	PSNR(dB)	29.81	31.95	32.02	31.99	32.54	32.42	32.01	32.40	32.86	32.77
		SSIM	0.8450	0.8850	0.8876	0.8843	0.8961	0.8944	0.8886	0.8931	0.9009	0.8993
		FSIM	0.8889	0.9274	0.9262	0.9271	0.9333	0.9305	0.9282	0.9297	0.9354	0.9353
		VIF	0.5246	0.5863	0.5927	0.5852	0.6088	0.6077	0.5933	0.6095	0.6219	0.6181
	×8	PSNR(dB)	25.96	27.34	27.60	27.75	28.20	28.07	27.60	28.26	28.40	28.43
		SSIM	0.7279	0.7519	0.7732	0.7749	0.7922	0.7890	0.7756	0.7957	0.7991	0.7997
		FSIM	0.8007	0.8586	0.8509	0.8597	0.8612	0.8556	0.8507	0.8593	0.8627	0.8639
		VIF	0.3418	0.3534	0.3869	0.3889	0.4195	0.4209	0.3925	0.4304	0.4349	0.4341
CelebA	×4	PSNR(dB)	27.69	29.67	29.24	29.76	29.86	30.08	29.97	30.00	30.24	31.09
		SSIM	0.8050	0.8506	0.8378	0.8540	0.8570	0.8699	0.8670	0.8680	0.8728	0.8809
		FSIM	0.8618	0.9005	0.8922	0.9027	0.9040	0.9113	0.9103	0.9097	0.9132	0.9209
		VIF	0.4369	0.5110	0.4899	0.5139	0.5209	0.5279	0.5192	0.5296	0.5318	0.5647
	×8	PSNR(dB)	24.16	26.29	26.04	26.12	26.48	26.35	26.13	26.49	26.69	26.84
		SSIM	0.6655	0.7404	0.7355	0.7339	0.7558	0.7527	0.7435	0.7616	0.7640	0.7635
		FSIM	0.7688	0.8463	0.8347	0.8444	0.8415	0.8375	0.8359	0.8420	0.8445	0.8471
		VIF	0.2696	0.3453	0.3367	0.3345	0.3650	0.3640	0.3464	0.3711	0.3780	0.3802

**Fig. 8.** Comparison of subjective results of different SR methods at scale of 8 on FFHQ. (a) LR input, (b) Bicubic, (c) DBPN (Haris et al., 2018), (d) RDN (Zhang, Tian, et al., 2018), (e) SAN (Dai et al., 2019), (f) RCAN (Zhang, Li et al., 2018), (g) SRFBN (Li et al., 2019), (h) MTC (Wang, Lu, et al., 2020), (i) SPARNet (Chen et al., 2021), (j) SISN (Lu et al., 2021), (k) SLNet, (l) Ground-truth. Local details are illustrated by the red boxes.

images that are visually pleasing, with rich textures and details, effectively improving the image blurring problem. That is, for SR images on the two datasets, our method generates very close images to the ground-truth images. To compare the subjective performance more intuitively, we show the local details of the reconstructed images in the bottom of each sample. Significantly, in the first row of Fig. 8 and the third row of Fig. 9, which visualize the eye and mouth parts of the face, respectively, our method obviously recovers better detail information. Specifically, according to Table 1, Figs. 8 and 9, we observe that the performance of face super-resolution-based methods (DBPN (Haris et al., 2018), RDN (Zhang, Tian, et al., 2018), SAN (Dai et al., 2019), RCAN (Zhang, Li et al., 2018)) is significantly better than that of image-super-resolution-based methods (SRFBN (Li et al., 2019), MTC (Wang, Lu, et al., 2020), SPARNet (Chen et al., 2021) and SISN (Lu et al., 2021)). Generally speaking, model-driven and data-driven are interdependent, and a particular model should be more suitable for the data of a particular scenario. Furthermore, the former uses more L_1 or L_2 loss functions, allowing

the generated images to be smooth. The latter mostly leverages the attention mechanism to make the model focus on detailed regions, leading to the generated images exhibiting sharp edges. In comparison, visual information fidelity (VIF) is more consistent with human intuition. SLNet's VIF indicator outperforms most comparison methods, as does the subjective visualization performance. As a result, the SR problem becomes challenging to face super-resolution, and multiple unknown degradation cases severely affect the reconstruction of high-quality HR images. In contrast, our approach can exploit complementary information between low and high-level spaces to recover missing details and thus achieve better SR performance.

4.3.3. Analysis of model complexity

We compare our method to several competitive methods (i.e., RDN (Zhang, Tian, et al., 2018), RCAN (Zhang, Li et al., 2018), SAN (Dai et al., 2019), SPARNet (Chen et al., 2021), SISN (Lu et al., 2021) in terms of the number of parameters (i.e., Params) and FLOPs. As shown in Tables 1 and 2, although our method has

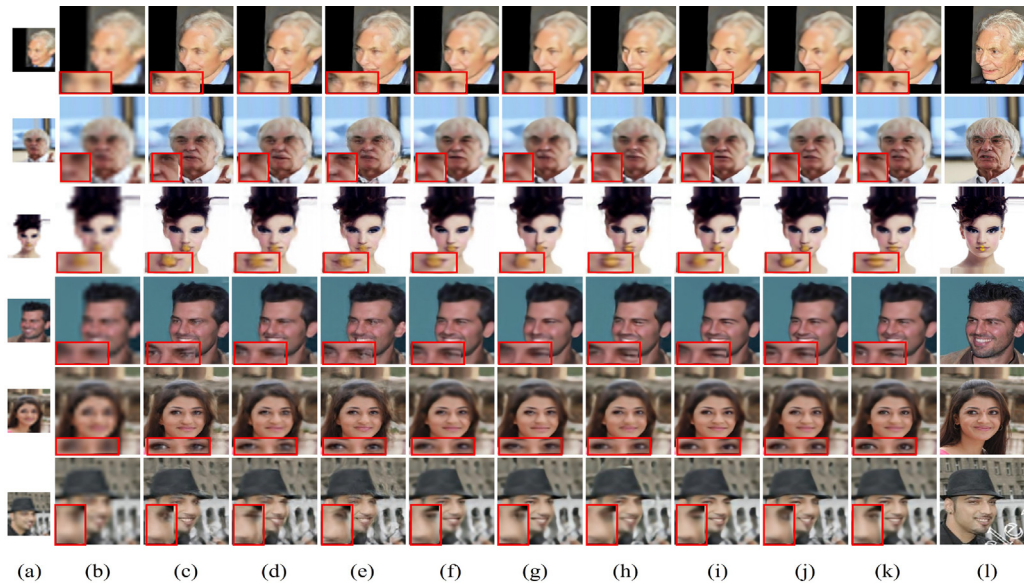


Fig. 9. Comparison of subjective results of different SR methods at scale of 8 on CelebA. (a) LR input, (b) Bicubic, (c) DBPN (Haris et al., 2018), (d) RDN (Zhang, Tian, et al., 2018), (e) SAN (Dai et al., 2019), (f) RCAN (Zhang, Li et al., 2018), (g) SRFBN (Li et al., 2019), (h) MTC (Wang, Lu, et al., 2020), (i) SPARNet (Chen et al., 2021), (j) SISN (Lu et al., 2021), (k) SLNet, (l) Ground-truth. Local details are illustrated by the red boxes.

Table 2
Comparison of the complexity of different methods for predicting SR images under scale $\times 4$ on CelebA dataset.

Method	RDN (Zhang, Tian, et al., 2018)	SAN (Dai et al., 2019)	RCAN (Zhang, Li et al., 2018)	SPARNet (Chen et al., 2021)	SISN (Lu et al., 2021)	SLNet
PSNR	29.24	29.76	29.86	30.00	30.24	31.09
SSIM	0.8378	0.8540	0.8570	0.8680	0.8728	0.8809
FLOPs	1.15	15.82	15.59	10.52	14.75	18.03
Params	3235	32562	31917	3638	21651	63919

a higher number of parameters and FLOPs than SPARNet (Chen et al., 2021), RCAN (Zhang, Li et al., 2018) SAN (Dai et al., 2019) and SISN (Lu et al., 2021), SLNet achieves competitive SR performance, especially under scale $\times 8$. Our method uses more complicated feature extraction and reconstruction models than theirs. These modules lead to performance improvements at the cost of reasonably increased parameters and FLOPs.

4.3.4. Analysis of the performance of real-world images

We compared our method with different methods on real-world LDHF (Maeng et al., 2012) datasets. Since it is challenging to acquire paired HR-LR images in real scenes, we used far images as input and compare the visual performance of different methods by using near images as reference HR images. Note that SLNet is tested using the CelebA trained model, in line with the other methods. Since the LR and reference HR images that are used as tests here do not achieve the absolute matching, it is not reasonable to measure objective quality metrics such as PSNR and SSIM.

The visualization results of the different methods are shown in Fig. 10. We can observe that all methods appear more or less blurred and even produce blending artifacts, mainly around the component regions of the face, such as the eyes, mouth, etc. This phenomenon may be because the deep learning-based methods aim to learn the nonlinear mapping relationship between HR and LR to complete the reconstruction. Moreover, most of the methods are good at improving the effect of noise on SR, but once there is not enough high-frequency information, then the reconstruction performance is not good. Nevertheless, SLNet achieves an impressive subjective performance, especially in the eye region. In conclusion, our method recovers finer details than SISN (Lu et al., 2021), RCAN (Zhang, Li et al., 2018) and SPARNet (Chen

et al., 2021) and produces fewer artifacts than MTC (Wang, Lu, et al., 2020), which indicates that our method can be applied to low-resolution face images to generate high-quality HR images.

To further validate the SR performance of different methods in the real world, we report two metrics for no reference image quality assessment, the average gradient (AG) (Wang, Du, Yu, & Wan, 2013) and image entropy (IE) (Tsai, Lee, & Matsuyama, 2008). Generally speaking, AG (i.e., image sharpness) reflects the ability of an image to express detailed contrasts, which is caused by significant differences in gray levels near the boundary or shadow line sides of the image. In an image, the rate of change of gray level in a certain direction is large, and its gradient is also large. Also, IE is expressed as the bit-average number of the set of image gray levels in bits/pixel, which also describes the average amount of information in the image. Therefore, it is reasonable to choose AG and IE to evaluate the quality of an image. It can be observed from Fig. 11 that SLNet and SISN exhibit the best image quality evaluation. We analyze the following factors: (1) there are differences in the subjective results of the different methods, showing structural smoothing and edge sharpness, which affect the AG and IE associated with the amount of information; (2) SISN and SLNet use deep split-attention network and a three-stage self-attention learning approach respectively to focus on relevant information of the face to yield high-quality SR images.

4.4. Ablation study

4.4.1. Self-attention module (SAM)

SAM module is used in our SLNet to self-supervise the learning of contextual information that facilitates reconstruction and to contact the information interaction of the LFSR and HFSR. We demonstrate its effectiveness by removing the proposed SAM

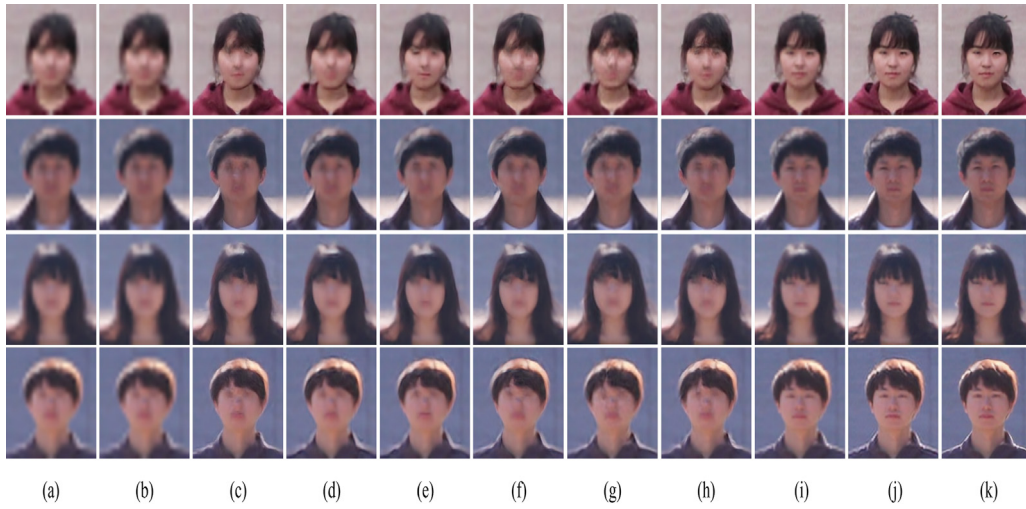


Fig. 10. Visualization of face SR results by different methods on LDHF. From left to right: (a) LR input, (b) Bicubic, (c) DBPN (Haris et al., 2018), (d) RDN (Zhang, Tian, et al., 2018), (e) SAN (Dai et al., 2019), (f) RCAN (Zhang, Li et al., 2018), (g) SRFBN (Li et al., 2019), (h) MTC (Wang, Lu, et al., 2020), (i) SPARNet (Chen et al., 2021), (j) SISN (Lu et al., 2021), (k) SLNet.

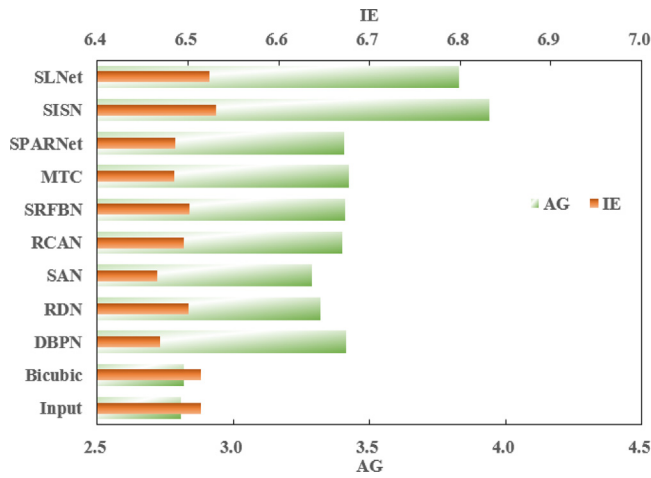


Fig. 11. Comparison of different methods for the no reference image quality assessment, including average gradient (AG) and image entropy (IE).

Table 3

Ablation study on the impact of SAM under scale of $\times 4$ on two datasets.

Dataset	Indicators	With SAM	Without SAM
FFHQ	PSNR	32.77	32.52
	SSIM	0.8993	0.8958
CelebA	PSNR	31.09	30.22
	SSIM	0.8809	0.8602

module from the final model. As shown in Table 3, for both publicly available datasets, the PSNR improves by 0.25 dB and 0.87 dB, SSIM improves by 0.0035 and 0.0207, respectively, when the SAM is removed. This is because SAM refines the feature extraction by fusing the original input image features with the output features of the first stage. Meanwhile, the attention mechanism strengthens the features with high attention and weakens the features with low attention, which is beneficial for the reconstruction of SLNet. In addition, SAM contributes equally to HFSR, as it feeds in features that favor SR for the continued propagation of information later.

We also show the feature maps of the output layers of different modules to evaluate the impact of SAM as well as LFSR and

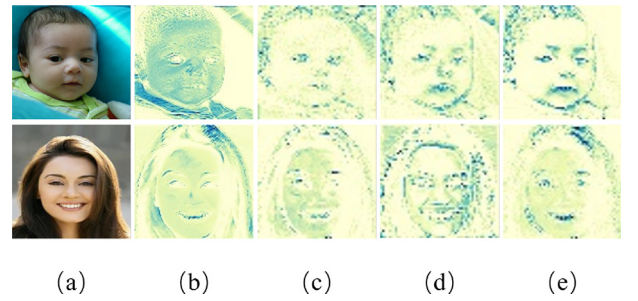


Fig. 12. Comparison of the output feature maps of different layers. From left to right: (a) Original input image, (b) Output feature map after LFSR, (c) Output feature map after SAM, (d) Output feature map of HFSR without SAM, (e) Output feature map of HFSR with SAM.

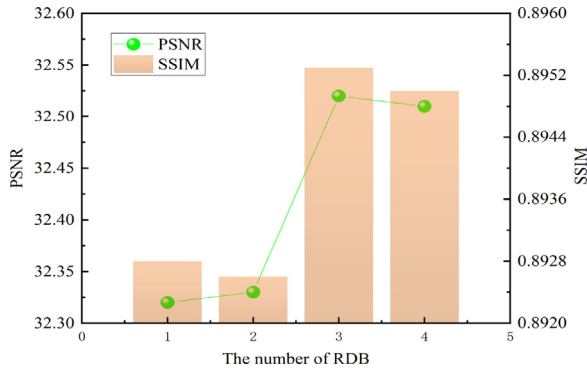
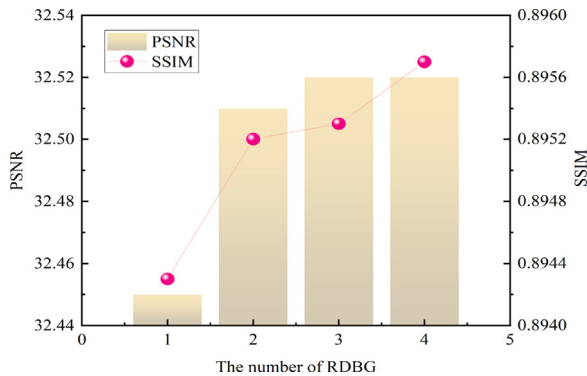
HFSR on the performance of SLNet. Fig. 12 visualizes the feature maps of the output of each layer for the two sample inputs as well as the original input maps. As we can see, the edge and texture components of the feature maps of SLNet (Fig. 12(e)) can be accurately located, while the other three feature maps appear poorly. Specifically, the output feature map of LFSR (Fig. 12(b)) shows very obvious blurring as a shallow feature with missing details. The output features after SAM (Fig. 12(c)) show better face contour information, but the detail information still cannot be accurately captured. Moreover, although the output feature map of HFSR without SAM (Fig. 12(d)) can enhance the edge and texture information, it also introduces redundant noise, such as the mouth and eye parts, which affects the whole quality of the image. In summary, the two-step reconstruction method with SAM designed by SLNet not only alleviates the redundant noise introduced during the deep feature extraction, but also enhances the feature representation on edges and textures to generate faithful image quality.

4.4.2. Design of the loss function

As mentioned in Section 3.5, SLNet uses three loss functions. For this purpose, four different cases are designed to determine the optimal combination of loss functions. The specific combinations of configurations and experimental results are shown in Table 4. Case A uses only the reconstruction losses that are usually used for SR. Cases B and C are primarily designed to verify the

Table 4
Ablation study with different combinations of loss functions.

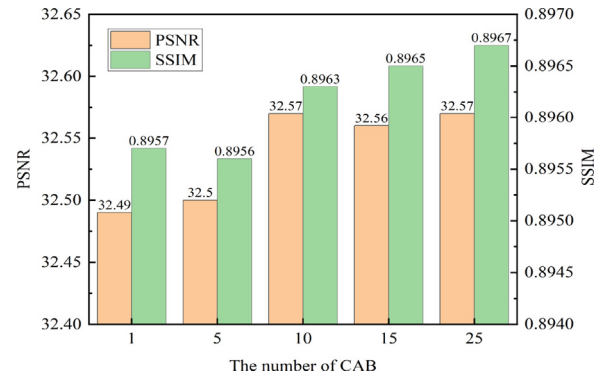
Case	L_{rec}	L_{sam}	L_{asy}	PSNR	SSIM
A	1	0	0	32.55	0.8961
B	1	1	0	32.57	0.8968
C	1	0	1	32.60	0.8971
D	1	0.5	0.5	32.69	0.8986

**Fig. 13.** Performance comparison on FFHQ using different numbers of RDB.**Fig. 14.** Performance comparison on FFHQ using different numbers of RDBGs.

importance of SAM and asymptotic loss. Experiments in cases D is designed to further determine the weighting factors for the three combinations. Compared to cases A, case B and case C show a 0.02 dB and 0.05 dB improvement in PSNR and 0.0007 and 0.0010 improvement in SSIM respectively. This indicates that SAM and pro are beneficial to SR and that using asymptotic loss is superior to SAM. Case D gives different weights for the three combined functions, which further improves the performance. L_{rec} serves as a direct constraint term to drive the generated image close to the ground truth, while L_{sam} and L_{asy} constrain both the intermediate result and the final reconstructed image in an indirect way. Following further analysis, the experiments for the four cases are finally set at 1, 0.5 and 0.5 for ω_1 , ω_2 and ω_3 respectively to achieve the best boost.

4.4.3. Number of RDBG

As the basic component unit in LFSR, RDBGs play a crucial role in feature extraction. Consequently, we investigate the SR performance with respect to the number of RDB and RDBG in our network. As can be seen in Figs. 13 and 14, the different number of RDB and RDBG show similar tendency in terms of performance. As the number of RDB and RDBG increases, the performance of the reconstruction consistently improves. However, the improvement tends to saturate when the number of RDB and RDBG increases from 3 to 4, respectively. Since the number

**Fig. 15.** Ablation study of the number of CAB on FFHQ ($\times 4$).

of parameters grows linearly with their number, we decided to use 3 RDBs (i.e. $K = 3$) and 3 RDBGs (i.e. $M = 3$) in SLNet to achieve a good trade-off between reconstruction accuracy and computational efficiency.

4.4.4. Number of CAB

We show the performance of using a different number of CABs in Fig. 15. These results indicate that larger G (the number of CABs) leads to higher performance, mainly due to the greater representativeness of the deeper network for larger G . On the other hand, larger G increases the number of parameters and decreases the algorithm's efficiency. Therefore, we use $G = 10$ in SLNet to achieve a trade-off between performance and efficiency.

5. Conclusion

In this paper, we propose a self-attention learning network that exploits a complementary three-stage FSR architecture and a simple self-attention module to explore information contributing to reconstruction to improve the degraded input gradually. On the one hand, LFSR and HFSR develop rules for feature learning that are independent but close-related to each other. It requires flexible information propagation between the complementary features in the two phases to enhance feature updating, thus boosting the collaboration between them. On the other hand, SAM establishes an attention-guided exchange of output information from the first stage to the third stage while incorporating features from both LFSR and HFSR. Our approach achieves competitive performance in facial super-resolution tasks. In the future, it is worthwhile to investigate in depth the use of deliberately designed attention in facial super-resolution, based on facial prior, perspective, pose, etc.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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